

Welcome to  
PHY 321:  
Classical  
Mechanics

The future that liberals want

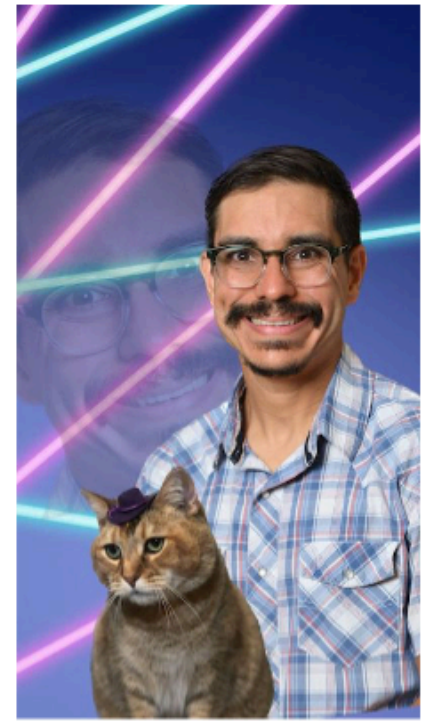


Prof. Danny Caballero (he/they; course instructor)  
Scarlett Rebolledo Caceres (she/her; GTA)  
Elisha Alemao (she/her; ULA)

# About Me



- Lappan Phillips Professor of Physics and Computational Science
- Co-direct two research labs (in Physics & Computational Science Education)
- PhD in Physics from Georgia Tech in 2011  
Postdoc CU-Boulder 2011-2013
- Former high school physics teacher  
(Atlanta Public Schools)
- Labor Organizer for Union of Tenure System Faculty
- Taught at MSU for 12 years.





# How to Contact Me

- Open "office" Hours
  - <https://cal.com/dannycaballero/phy321>
  - Zoom (in person if requested)
- Office hours (Physics Help Room; Strosacker Ctr)
  - MW 4-5; F 3-4; or by appointment!
- Response Hours: 5am-4pm, M-F
  - We all work different hours
- Response time: 1-2 business days
  - After 2 days, please send me another email/message!
- Email (caball14@msu.edu)



# Getting to know you!

- Student Information Survey is on website (Getting Started) and on D2L
  - Name & Pronouns
  - Python Experience
  - Access needs
- Fill this short survey out by **11:59pm Friday, January 17th**
- Why is learning about you important?
  - We care about you as an individual.
  - We should know what you need/want from this course.
  - We learn better in a community that supports us.

# Important Links (login with MSU credentials)

- Course website:  
<https://dannycab.github.io/phy321msu/intro.html>
- D2L – <https://d2l.msu.edu/d2l/home/2127041>
- Gradescope – <https://www.gradescope.com/courses/944711>  
(you will need a scanning device such as a phone camera, computer/tablet camera, etc.)
- **Please check that you can access D2L and that you've been added to the course on Gradescope.**

**Contact Danny if you have any issues!**

# Course Description

- This course is designed to help you to learn classical physics concepts using both analytical and computational methods. Through examination of the fundamental principles of classical mechanics, we will gain a deeper understanding of how objects and systems move.
- **Recommended Textbooks**
  - (JRT): Taylor, J.R. (2005). *Classical mechanics*. University Science Books.
  - (AMS): Malthe-Sørenssen, A. (2015). *Elementary Mechanics Using Python: A Modern Course Combining Analytical and Numerical Techniques*. Springer.
- **Supplementary Textbooks**
  - (MLB): Boas, M. L. (2006). *Mathematical methods in the physical sciences*. John Wiley & Sons.

**See "Textbooks" folder on D2L for the library catalog resource.**

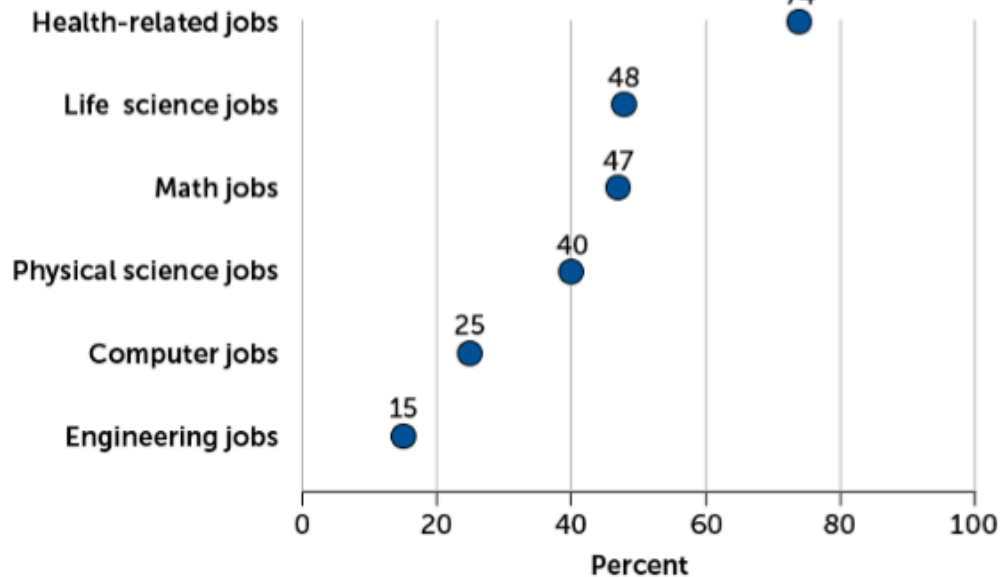
# Inclusive Classrooms

- **Inclusive classroom:** Where instructors and students work together to create and sustain an environment in which everyone feels safe, supported, and encouraged to express their views and concerns (UMich CTL definition)

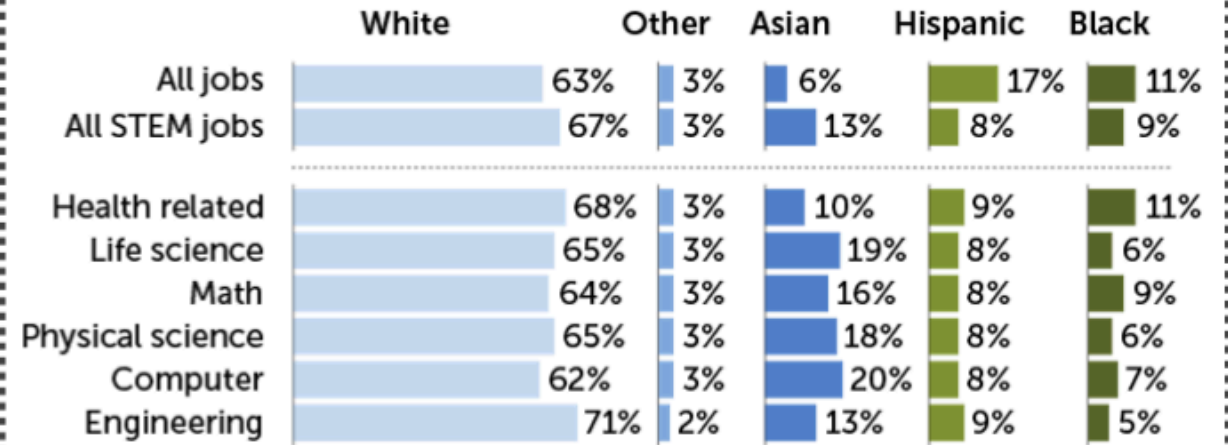


# Racial and Gender Gaps Persist in STEM

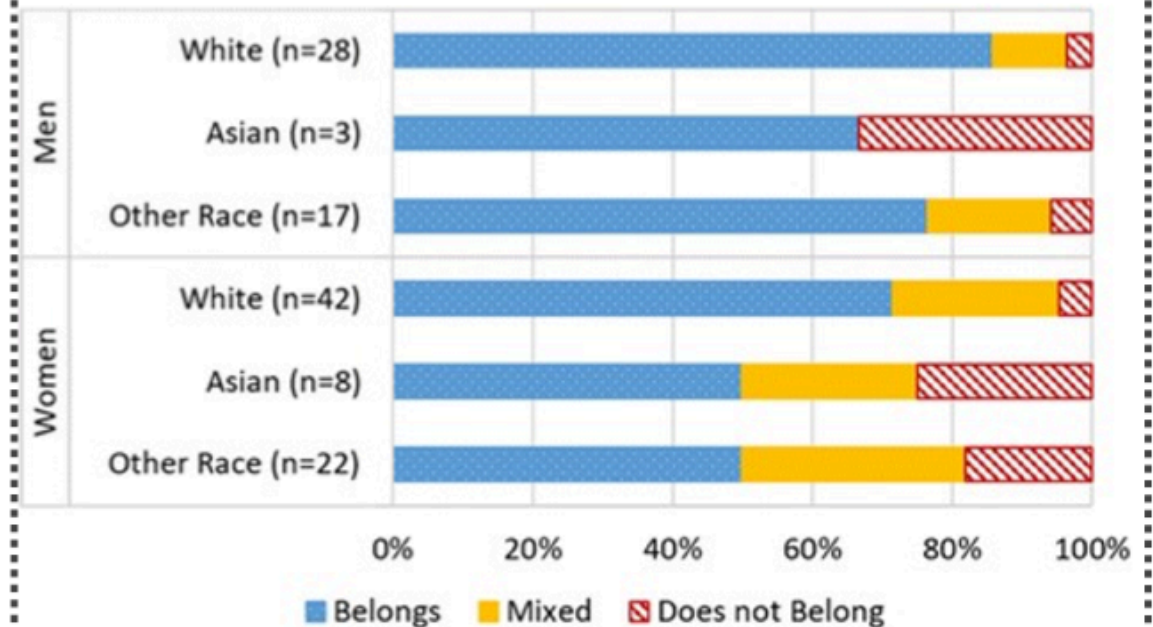
**Percentage of STEM professionals who are women (by field), 2017-2019**



**Racial and Ethnic representation in STEM jobs (2017-2019)**



**Feelings of Belonging in STEM (by gender and race)**





# Inclusive Classrooms

- **Inclusive classroom:** Where instructors and students work together to create and sustain an environment in which everyone feels safe, supported, and encouraged to express their views and concerns (UMich CTL definition)
- We strive to create inclusive classroom spaces, but we can't do it without you
- You will need to work together to solve the problems we present while making sure that everyone has the chance to contribute to the effort
- **Issues or concerns:** Please bring those to Danny. If you prefer, you can also contact our associate chair, Prof. Stuart Tessmer, or, if needed, the department chair, Prof. Steve Zepf.

# Course Activities

- Homework (15%) – 9 of them DUE Fridays at 11:59pm ET **via Gradescope**
  - Problem Sessions on Fridays
  - Can work/submit in small groups (2-3 people) OR individually
  - Each part graded on a 3-point rubric (correct, slightly incorrect, incorrect, blank)
- Individual Reflections (5%) -- 9 of them DUE Fridays at 11:59pm ET **via D2L**
  - Designed to encourage you to think critically about your learning process and identify areas for growth
  - Prompts to help you reflect on your experience in 1-2 paragraphs

# Learning is a social and collaborative act!

- It's an essential skill in science and engineering (and highly valued by employers)!
- Social interactions are critical to scientists' success – most good ideas grow out of discussions with colleagues, and essentially all physicists work as part of a group.
- **For HW, you can work in small groups (2-3 people) OR by yourself. If you work as a group, you can turn in the assignment as a group – REMEMBER TO WRITE YOUR NAME(S) AND ADD YOUR GROUP MEMBERS TO THE GRADESCOPE SUBMISSION!**



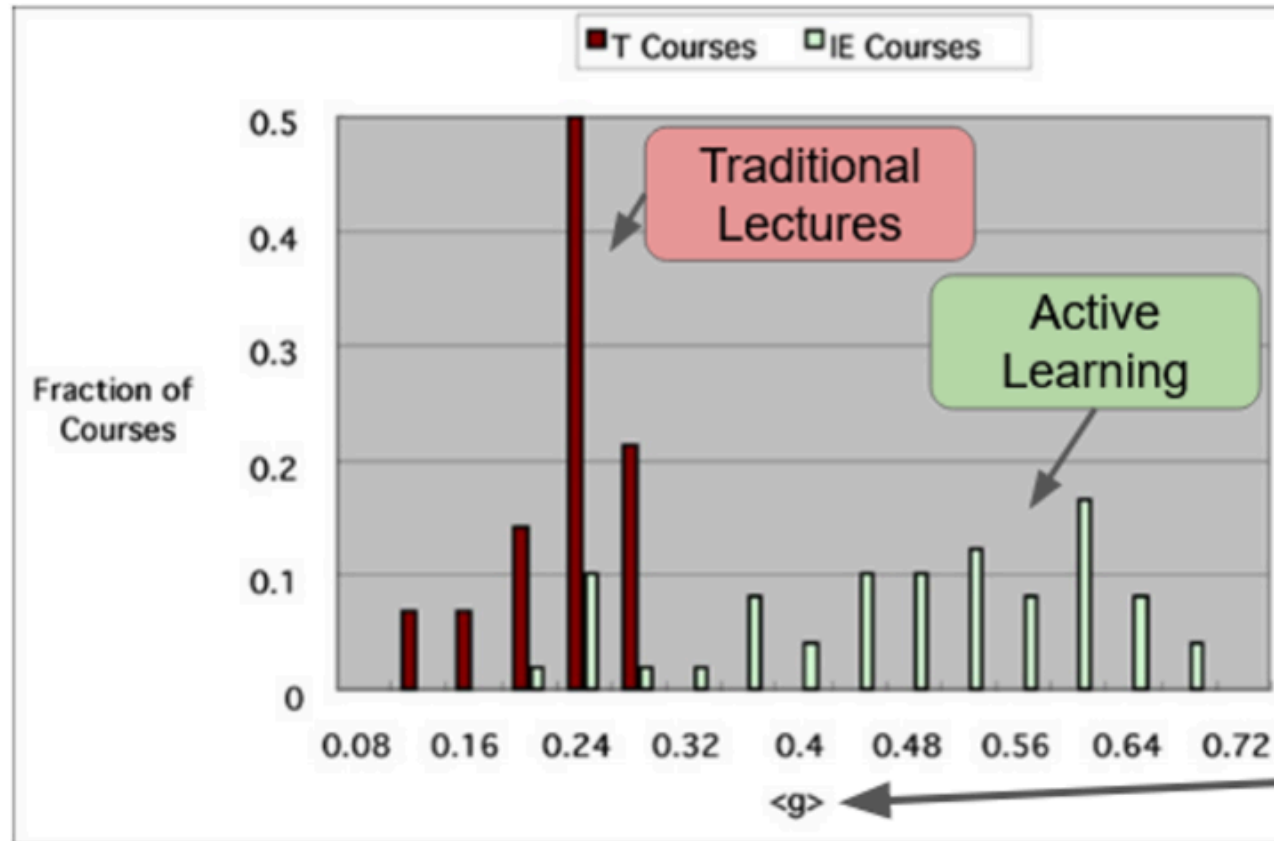
# Course Activities

- Midterm Projects (50%) -- 2 of them DUE Feb 28<sup>th</sup> and Apr 11<sup>th</sup> at 11:59pm **via Gradescope**
  - Will look like the Homework but slightly beyond what you've already done
  - **Must work/submit individually**; you can work together and with us
  - Encouraged to use the textbook, notes, and other resources when solving the midterms and you can ask any of the teaching staff for help
- Final Exam (30%) -- DUE Friday, May 2<sup>nd</sup> at 11:59pm **via D2L**
  - Analyze a physical system of your choice and prepare a narrative using Jupyter notebooks
  - Can work in small groups (2-3 people) OR individually
  - **Must submit individually**

# Extra Credit Opportunities

- Using iClickers during Class
  - Used in class to gauge your understanding of a topic or concept.
  - Will not be penalized for not knowing the correct answer.
  - Earn up to **1% extra credit to your overall homework grade**
- Attending the Department of Physics & Astronomy Seminar and/or Colloquia
  - Summarize the talk using at least 150 words and turn in your summary along with your homework assignment.
  - Earn up to **5 extra credit points on your homework assignment**
- Completing "Challenge Assignments" on D2L (posted soon)
  - These challenges (5 of them) are designed to aid you in developing a set of skills as you continue to advance in your career.
  - Earn up to **2% extra credit to your overall grade**

# Measuring Student Learning



Larger is better

R. Hake, "...A six-thousand-student survey..."  
*AJP* 66, 64-74 ('98).

$$\langle g \rangle = \frac{post - pre}{100 - pre}$$

# Generative AI Policy for PHY 321

- On Friday, we will work together to develop a policy for our class.
- In preparing for that conversation, **please take at least 30 minutes to review the following link.**
  - <https://openpraxis.org/articles/10.55982/openpraxis.16.4.777>
- If you want more information, there are podcast episodes, videos, etc. posted on course website and D2L as well.
- Ask yourself:
  - What forms of AI use are acceptable to me? Why do I think that?
  - How would I demonstrate that I understand this material (i.e., your work is not just the product of AI)?
  - What forms of documentation should be used to indicate that I used AI tools?
  - When is it unacceptable to me to use AI in this class? Why do I think that?
  - What should happen if someone using AI in this class in a way that I find inappropriate?



# Accommodations

If you have a university-documented learning difficulty or require other accommodations, please provide me with your VISA as soon as possible and speak with me about how I can assist you in your learning.

If you do not have a VISA but have been documented with a learning difficulty or other problems for which you may still require accommodation, please contact MSU's Resource Center for People with Disabilities (355-9642) in order to acquire current documentation.

If you need accommodations but do not yet have documentation, please contact me!



**Questions?**

# Think, Pair, Share Activity

- *What are 1-2 things you are excited for this semester?*
- *What are 1-2 things you are nervous/anxious about this semester?*
  - *What questions (if any) are still lingering for you?*
- *Think:* Take 4 minutes (I'll set a timer) to document your answers to these questions above
- *Pair:* Discuss your answers with the people around you (5 min).
- *Share:* Turn in your answers (you don't have to put your name on it but you can if you'd like me to know)

# Software Set Up

- Double check that you have access to the **D2L** and **Gradescope** webpages
- Download **Python** via the Anaconda distribution (see the "Start Here" folder on D2L for link)
- Set up your **iClicker** account on your smartphone, tablet or laptop (see the "Start Here" folder on D2L for directions)
- Fill out the **Student Information Survey** (see "Start Here" folder on D2L for link)



# Next Class (Wednesday 1/15)

- Read **Ch. 3.4** from Mathematical Methods in the Physical Sciences (MLB)
  - MSU library has electronic version (see link on D2L)
  - Download and install Anaconda Python
- Discussion about "Classical Mechanics" and vectors